

BANK ANALYTICS

Debit & Credit Analysis

Using MySQL · Excel · Power BI · Tableau

Developed by: Team

Database: Bank_Analytics

Group Details

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Project Summary

OBJ

Objective

Build reports and dashboards that track loan trends, repayment performance, and portfolio risk — and analyse transaction patterns and customer behaviour across debit and credit activity, branch operations, and transaction methods.

DB

Database

Structured MySQL database under a banking data schema, integrating loan portfolio records and customer transaction data across branches.

TOOL

Tools Used

MySQL | Excel | Power BI | Tableau — each used for progressive layers of analytics and visualisation.

IMPACT

Business Impact

Identified branch performance, default & delinquency risk, and transaction trends to strengthen lending strategy, collections, and fraud detection.

Bank Loan Analytics — KPIs

1

Total Loan Amount Funded

Total value of loans disbursed to clients

2

Total Loans

Number of loans issued, active and closed

3

Total Collection

Principal and interest collected from borrowers

4

Total Interest

Revenue generated from loan interest

5

Branch-Wise Revenue

Interest, fees & total revenue by branch

6

State-Wise Loan

Geographic distribution of loans by state

7

Product Group-Wise Loan

Loan volume by product type — personal, auto, etc.

8

Disbursement Trend

Loan disbursement trend over time

9

Grade-Wise Loan Risk

Portfolio risk by borrower credit grade

10

Default & Delinquent Rate

% of loans overdue or in default

11

Loan Status-Wise Loan

Breakdown by active, closed, delinquent status

12

Age Group-Wise Loan

Loan allocation by borrower age group

Debit & Credit Analysis — KPIs

1

Total Credit Amount

Sum of amount where transaction type = Credit

2

Total Debit Amount

Sum of amount where transaction type = Debit

3

Credit to Debit Ratio

Total credit ÷ total debit amount

4

Net Transaction Amount

Total credit minus total debit (net cash flow)

5

Account Activity Ratio

Number of transactions ÷ account balance

6

Transactions per Day/Week/Month

Transaction volume trend over time

7

Transaction Amount by Branch

Total transaction volume grouped by branch

8

Transaction Volume by Bank

Transaction activity compared across banks

9

Transaction Method Distribution

Share of transfer, debit card, credit card use

10

Branch Transaction Growth

% change in branch transaction volume over time

11

High-Risk Transaction Flag

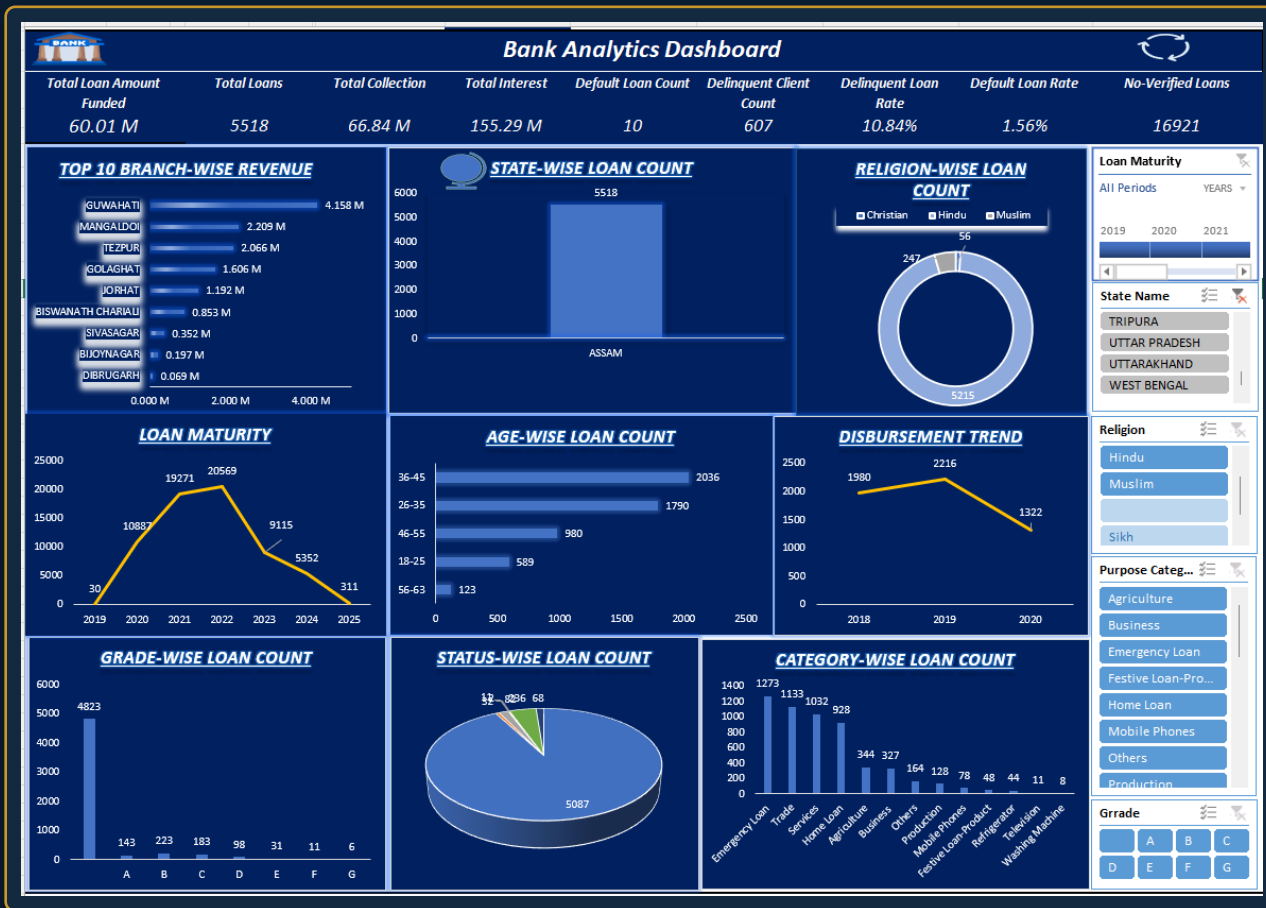
Transactions exceeding threshold or irregular

12

Suspicious Transaction Frequency

Count of flagged high-risk transactions

Excel Dashboard — Bank Loan Analytics



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BANK TRANSACTION ANALYSIS DASHBOARD

Total Credit
₹127.603M

Total Debit
₹127.603M

Credit/Debit Ratio
1.00

Net Transaction
318.1K

Activity Ratio
19.01

Suspicious Txn Frequency
20.4K

Transaction Method Distribution

Bank Transfer

Credit Card

Debit Card

33352

33338

33310

High Risk Transaction Flag

10196

High Risk Deposit

10230

High Risk Withdrawal

79574

Normal

Transaction Volume By Bank

Year

2024

Quarter

Q1

Q2

Q3

Q4

Kotak Mahindra Bank

Axis Bank

State Bank of India

ICICI Bank

Punjab National Bank

HDFC Bank

Total Txn Amount by Branch

City Center Branch

Main Branch

East Branch

Downtown Branch

Suburban Branch

North Branch

Monthly Transaction Trend

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

Dec

Transaction Type

Credit

Debit

Bank Name

Axis Bank

HDFC Bank

ICICI Bank

Kotak Mahindra Bank

Punjab National Bank

State Bank of India

Branch

City Center Branch

Downtown Branch

East Branch

Main Branch

North Branch

Suburban Branch

BRANCH Growth Analysis Report

MonthName

January

February

March

April

May

June

July

August

September

October

November

December

Tableau Dashboard — Bank Loan Analytics (Overview)

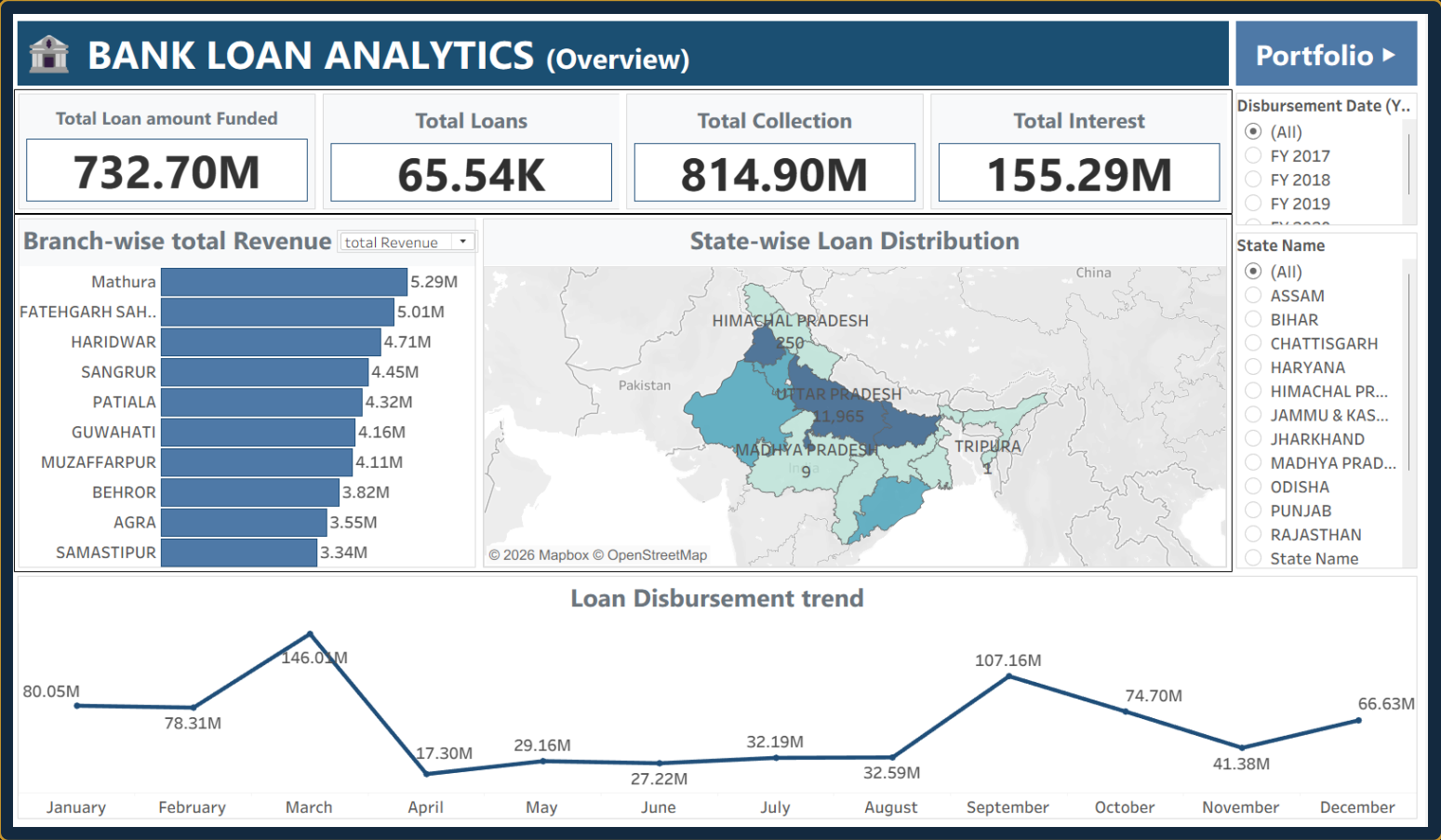


Tableau Dashboard — Loan Portfolio & Risk Analysis

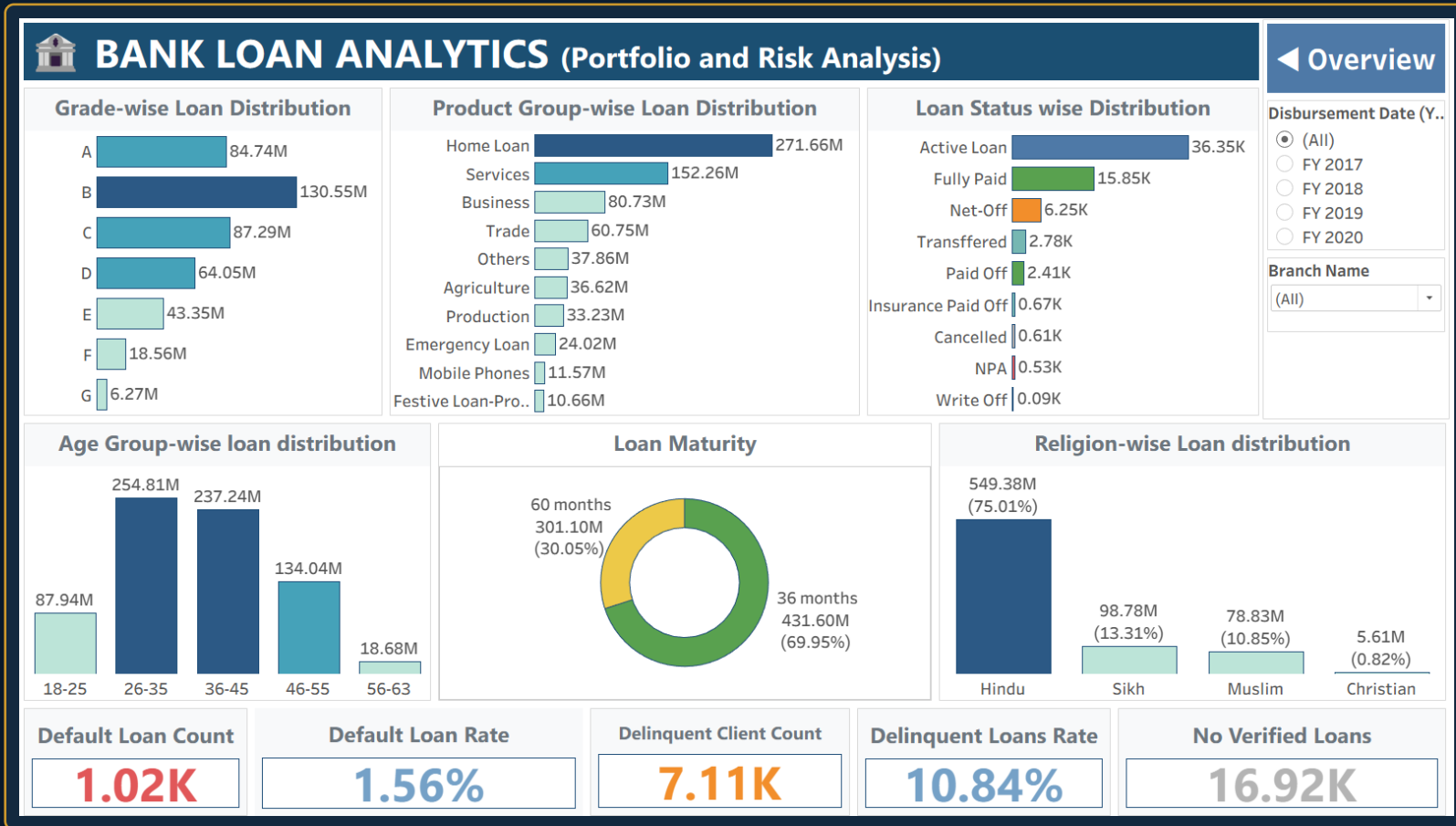
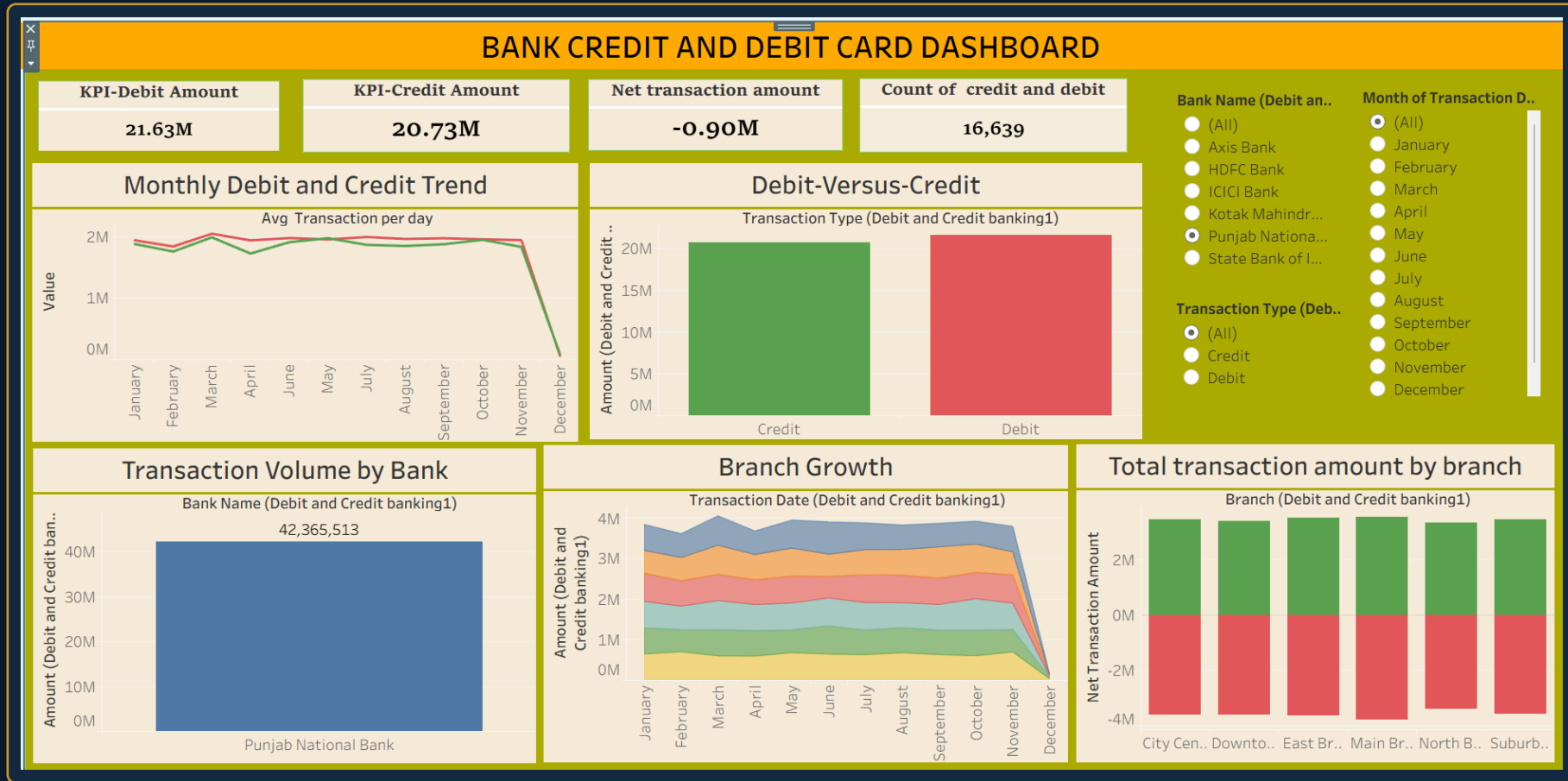


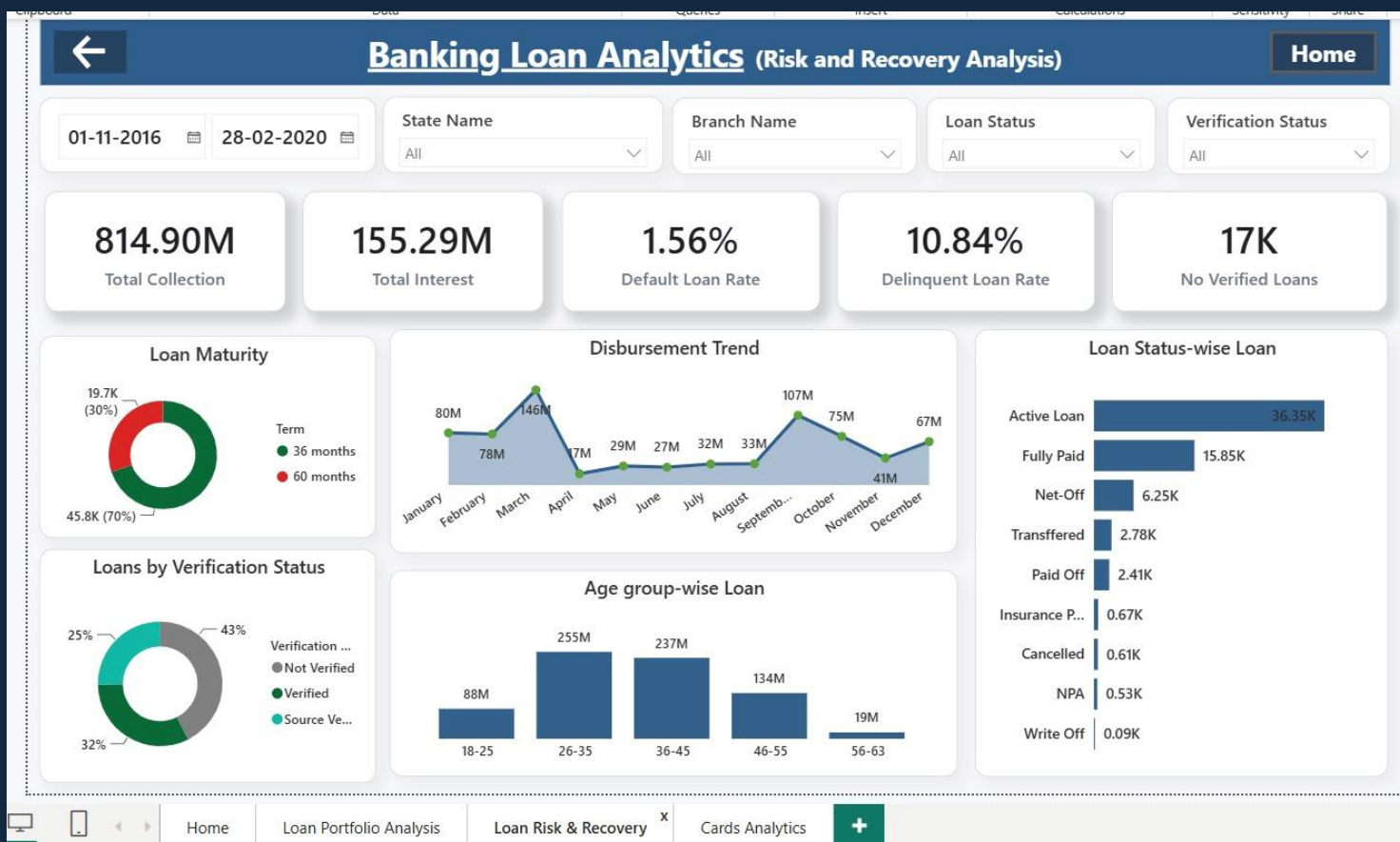
Tableau Dashboard — Debit & Credit Analysis



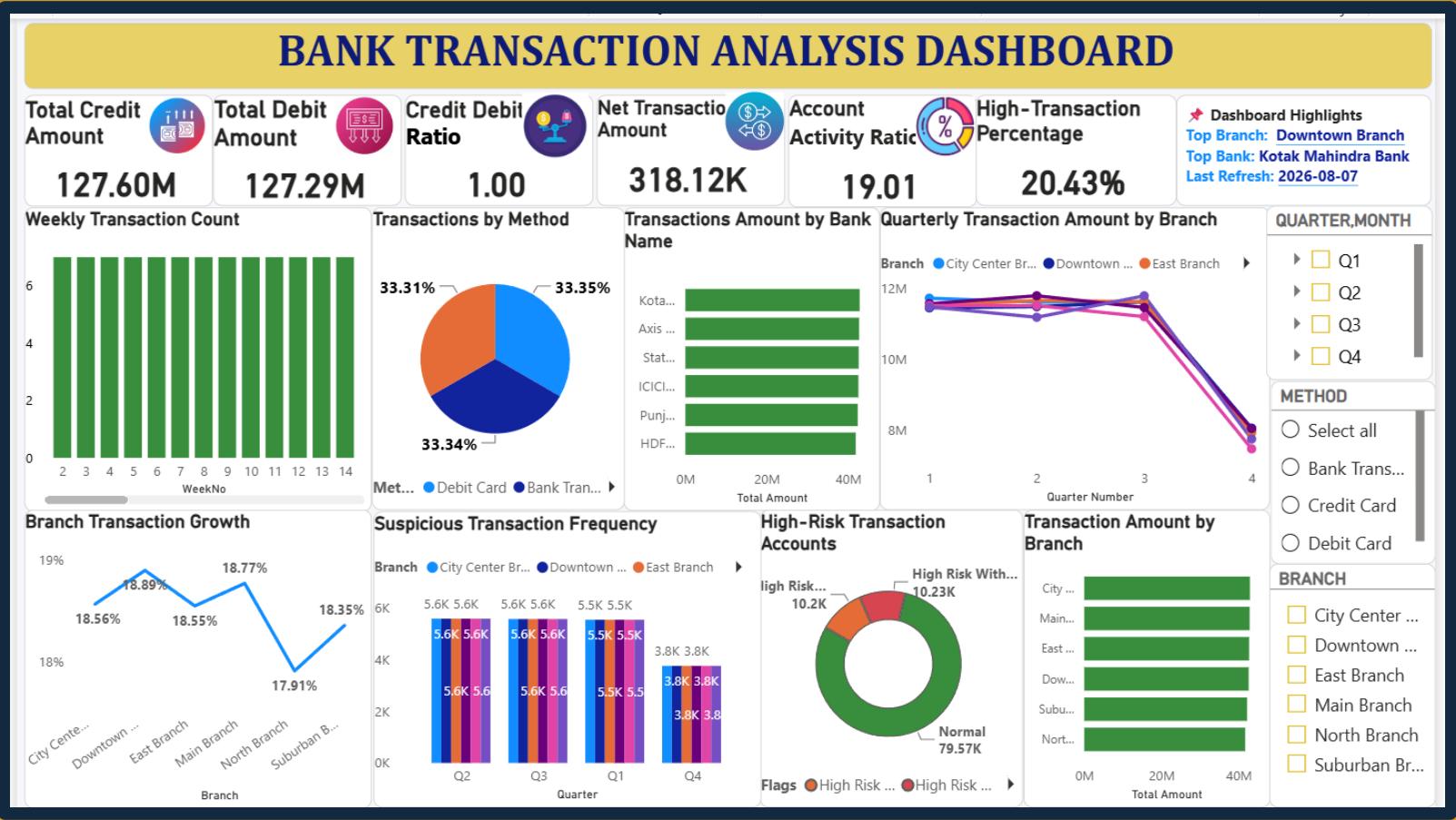
Power BI Dashboard — Bank Loan Analytics



Power BI Dashboard — Loan Risk & Recovery Analysis



Power BI Dashboard — Debit & Credit Analysis



SQL Query — Bank Loan Analytics

Content to be added

SQL Query — Debit & Credit Analysis

Content to be added

Challenges Faced

1

Clean Source Data

Loan & transaction datasets arrived largely clean, so cleaning effort was minimal.

2

Derived Fields in Cards Data

An extra computed column had to be built with custom formulas; AI tools helped clarify terminology and identify the right source columns.

3

SQL Import Issues

Direct file import failed for some tables, so data had to be loaded using custom SQL queries instead.

4

Power BI KPI Selection

Early uncertainty over which measures to include, and whether to show consolidated vs. per-account figures.

5

Cross-Tool Consistency

Keeping metric definitions & visual design aligned across Excel, SQL, Power BI and Tableau needed extra coordination.

6

Team Coordination

Synchronizing work across a 6-member team on different dashboard sections to avoid conflicting logic or duplicated effort.

Key Takeaways

1

Branch performance has a wide spread

Mathura, Fatehgarh Sahib and Haridwar bring in the most loan revenue — a few branches near the bottom of the list are worth a closer look.

2

Loan risk is real, but manageable

Only 1.56% of loans default and about 10.84% are delinquent — a small slice of the book, but enough to keep watching closely.

3

Money in, money out — almost perfectly balanced

Total credit and debit amounts land within a hair of each other, with a credit-to-debit ratio sitting right around 1.00.

4

A small set of transactions carries most of the risk

Roughly 20K transactions get flagged as high-risk or suspicious — a small share overall, but exactly where fraud monitoring should focus.

5

The cards dataset was the real challenge

Building it end to end across Excel, SQL, Power BI and Tableau taught us as much about teamwork as it did about the numbers.

Thank You!

Bank Analytics — Debit & Credit Analysis

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